

Project

Intelligent Network Monitoring System Using Device Metrics

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Project Overview

Modern organizations rely heavily on computer networks to support business operations. As networks continue to grow in size and complexity, manually monitoring network devices becomes inefficient and increases the risk of undetected failures, performance degradation, and prolonged downtime.

This project proposes the development of an **Intelligent Network Monitoring System** that continuously collects performance metrics from network devices, analyzes their health, and presents the information through an interactive web dashboard. The system enables administrators to monitor network performance in real time, visualize the network topology, receive automated alerts, and identify potential issues before they impact users.

The objective is to improve network reliability, reduce troubleshooting time, and provide centralized visibility into the organization's infrastructure.

Problem Statement

Many organizations still depend on manual monitoring methods or basic tools that provide limited visibility into the health of network devices.

Some of the common challenges include:

- Lack of centralized monitoring of network devices.
- Difficulty identifying devices experiencing high CPU utilization.
- Delayed detection of network failures.
- No real-time severity classification of device health.
- Limited visualization of network topology.
- Increased downtime due to slow incident response.

These challenges reduce network availability and increase operational costs.

Proposed Solution

The proposed Network Monitoring System automatically gathers metrics from routers, switches, servers, and other network devices.

The collected metrics are analyzed and displayed through an intuitive dashboard that provides:

- Real-time monitoring of device health.
- CPU utilization monitoring.
- Device status monitoring (Online/Offline).

- Interactive network topology visualization.
- Analytics and historical trends.
- Automatic alert generation.
- Severity classification of detected issues.
- Centralized management interface.

The system allows network administrators to quickly identify problems and respond before they become critical.

Project Objectives

Main Objective

To develop a centralized web-based Network Monitoring System that continuously monitors device metrics and provides real-time analytics, topology visualization, and intelligent alerts.

Specific Objectives

- Monitor CPU utilization of network devices.
- Display device status in real time.
- Generate alerts when device thresholds are exceeded.
- Classify alerts based on severity.
- Display historical CPU performance.
- Visualize network topology.
- Provide network performance analytics.
- Reduce troubleshooting time.
- Improve network availability.

System Features

1. Dashboard

The dashboard acts as the control center of the monitoring system.

It displays:

- Total devices
- Online devices
- Offline devices
- Active alerts
- CPU utilization summary

- Recent network events
- Historical performance graphs

Benefits:

- Single-pane visibility
- Real-time updates
- Easy monitoring

2. Device Monitoring

Each registered device is monitored continuously.

Metrics collected include:

- Device name
- IP address
- Device type
- CPU utilization
- Device status
- Last updated timestamp

The dashboard refreshes automatically as new metrics are received.

3. CPU Monitoring

CPU utilization is continuously monitored to detect abnormal behavior.

Example thresholds:

CPU Usage	Severity	Status
0–49%	Low	Healthy
50–69%	Medium	Warning
70–89%	High	Critical
90–100%	Very High	Emergency

This allows administrators to prioritize incidents based on urgency.

4. Alert Management

Whenever CPU utilization exceeds predefined thresholds, the system automatically creates alerts.

Each alert contains:

- Device name
- CPU usage
- Timestamp
- Severity
- Alert message

Example:

Switch-01

CPU Usage: 92%

Severity: Emergency

Message:

"CPU utilization has exceeded the critical threshold."

5. Severity Classification

The monitoring system classifies events into four levels.

Low

- Normal operation
- No administrator action required

Medium

- Performance beginning to degrade
- Administrator should investigate

High

- Significant performance issue
- Immediate attention recommended

Critical

- Device at risk of failure
- Immediate intervention required

This classification helps prioritize responses.

6. Network Analytics

The dashboard provides analytics including:

- CPU utilization trends
- Average CPU usage
- Device availability
- Online vs Offline devices
- Alert frequency
- Device performance history

Graphs help administrators identify recurring issues before they become outages.

7. Network Topology Visualization

The system displays a graphical representation of connected devices.

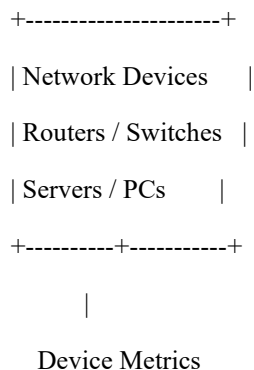
Topology includes:

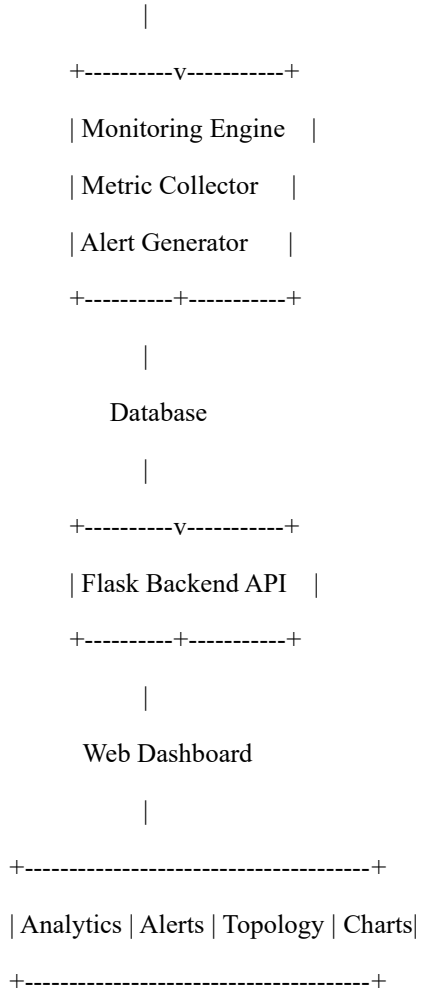
- Routers
- Switches
- Servers
- PCs
- Connections between devices

Benefits include:

- Easier fault isolation
- Better understanding of network structure
- Faster troubleshooting

System Architecture





Workflow

1. Devices generate performance metrics.
2. Monitoring engine collects CPU utilization.
3. Metrics are stored in the database.
4. Dashboard retrieves updated metrics.
5. Charts display historical performance.
6. Alert engine checks thresholds.
7. Alerts are generated if thresholds are exceeded.
8. Severity level is assigned.
9. Administrator receives visual notification.

Benefits of the System

The proposed solution offers several benefits:

- Centralized monitoring
- Reduced network downtime
- Faster incident response
- Early detection of device failures
- Improved decision making
- Better resource utilization
- Historical performance tracking
- Scalable architecture
- Easy deployment
- User-friendly dashboard

Future Improvements

Future enhancements may include:

- Email notifications
- SMS alerts
- Machine Learning for anomaly detection
- Predictive maintenance
- Bandwidth monitoring
- Memory utilization monitoring
- Disk usage monitoring
- Mobile application
- Multi-site monitoring
- Role-based access control

Expected Results

After implementation, the system is expected to:

- Monitor all registered devices in real time.
- Display accurate CPU utilization.
- Provide real-time alerts.
- Categorize issues by severity.
- Visualize the network topology.

- Improve overall network availability.
- Reduce troubleshooting time.
- Increase operational efficiency.

Conclusion

The Intelligent Network Monitoring System provides a modern approach to managing network infrastructure by combining real-time monitoring, analytics, topology visualization, and automated alerting into a single web-based platform. By continuously collecting device metrics and presenting actionable insights through an intuitive dashboard, the system enables administrators to detect issues early, prioritize incidents using severity levels, and maintain optimal network performance.

This project demonstrates how data-driven network monitoring can enhance operational efficiency, reduce downtime, and support proactive network management. It provides a scalable foundation that can be expanded with advanced capabilities such as SNMP integration, predictive analytics, and automated notifications, making it a valuable solution for organizations seeking to improve the reliability and resilience of their network infrastructure.